

Song Liao

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RESEARCH INTERESTS

- Policy and Privacy Compliance on Open Platforms
- IoT Security and Privacy
- Abuse Detection on Online Social Platforms

PROFESSIONAL EXPERIENCE

Assistant Professor (Tenure-Track) 08/2024 - Present
Texas Tech University, Lubbock, TX

EDUCATION

Clemson University, Clemson, SC 08/2019 - 08/2024

- Ph.D. in Computer Science
- Advisor: Dr. Long Cheng

Xi'an Jiaotong University, Xi'an, China 09/2015 - 06/2018

- M.E. in Software Engineering
- Advisor: Dr. Yuehu Liu

Xi'an Jiaotong University, Xi'an, China 09/2011 - 06/2015

- B.E. in Software Engineering

GRANTS

Collaborative Research: CyberAI Innovation: Securing and Leveraging Agentic AI for Cyber Defense
Sponsoring Agency: NSF 2026 - 2029

- Status: Granted (PI)
- Award Amount: \$249,848 (total: \$500,000)
- This project will develop an AI agent-centered cybersecurity education framework organized around curriculum modules, hands-on labs and project-based learning.

PUBLICATIONS

(Students advised by me, * denotes equal contribution.)

- [1] Weiliang Chen, **Song Liao**, “RATED: Reliability-Aware AI Agent Tool Retrieval via Execution Dynamics”, *ACM International Conference on Information and Knowledge Management (CIKM)*, 2026, Accept rate: $236/764 = 31\%$.
- [2] Huan Zhao, Kan Yang, **Song Liao**, “Diffusion-Aware Error Correction Code Watermarking Framework”, *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2026.
- [3] Felix Sarpong, **Song Liao**, “Adversarial Attacks on Differentially Private Factorization Machines in Recommendation Systems”, *ACM Symposium on Access Control Models and Technologies (SACMAT)*, 2026, Accept rate: $16/56 = 29\%$.
- [4] John Anderson, Mohammed Aldeen, Shuhao Zhang, **Song Liao**, Hongxin Hu, Mashrur Chowdhury, Long Cheng, “Zero Trust Security Technologies, Applications, and Adoption Challenges”, *EAI International Conference on Security and Privacy in Cyber-Physical Systems and Smart Vehicles (SmartSP)*, 2025.
- [5] Andrew York, Mohammed Aldeen, Jingwen Yan, Pranav Silimkhan, **Song Liao**, Mert D Pesé, Long Cheng, “A First Look at Privacy Compliance of Zoom Apps”, *IEEE Secure Development Conference (SecDev)*, 2025.
- [6] Ebuka Okpala, Nishant Vishwamitra, Keyan Guo, **Song Liao**, Long Cheng, Hongxin Hu, Yongkai Wu, Xiaohong Yuan, Jeannette Wade, Sajad Khorsandroo, “AI-Cybersecurity Education Through Designing AI-based Cyberharassment Detection Lab”, *Journal of The Colloquium for Information Systems Security Education*. Vol. 12. No. 1. 2025.
- [7] Nishant Vishwamitra, Ebuka Okpala, **Song Liao**, Keyan Guo, Sandeep Shah, Hongxin Hu, Xiaohong Yuan, Long Cheng, “Enhancing AI-Centered Social Cybersecurity Education through Learning Platform Design”, *Journal of The Colloquium for Information Systems Security Education*. Vol. 12. No. 1. 2025.
- [8] Jingwen Yan, **Song Liao**, Jin Ma, Mohammed Aldeen, Salish Kumar, Long Cheng, “No Way to Sign Out? Unpacking Non-Compliance with Google Play’s App Account Deletion Requirements”, *USENIX Security Symposium (USENIX Security)*, 2025, Acceptance rate: $407/2385 = 17\%$.

- [9] Jingwen Yan*, **Song Liao***, Mohammed Aldeen, Luyi Xing, Danfeng (Daphne) Yao, Long Cheng, “SkillPoV: Towards Accessible and Effective Privacy Notice for Amazon Alexa skills”, *Network and Distributed System Security Symposium (NDSS)*, 2025, Acceptance rate: $211/1311 = 16\%$.
- [10] Shuhao Zhang, Mohammed Aldeen, **Song Liao**, Jeffery Young, Long Cheng, “AcousticScope: Understanding Biases in Voice Interaction via Automated Acoustic Testing”, *EAI International Conference on Security and Privacy in Cyber-Physical Systems and Smart Vehicles (SmartSP)*, 2024. (Best Demo Award)
- [11] Mohammed Aldeen, Jeffrey Young, **Song Liao**, Tsu-Yao Chang, Long Cheng, Haipeng Cai, Xiapu Luo, Hongxin Hu, “End-Users Know Best: Identifying Undesired Behavior of Alexa Skills Through User Review Analysis”, *ACM Interactive, Mobile, Wearable, and Ubiquitous Technologies (UbiComp)*, 2024, Acceptance rate: 23%.
- [12] **Song Liao**, Long Cheng, Xiapu Luo, Zheng Song, Haipeng Cai, Danfeng (Daphne) Yao, Hongxin Hu, “A First Look at Security and Privacy Risks in the RapidAPI Ecosystem”, *ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2024, Acceptance rate: $331/1964 = 17\%$.
- [13] **Song Liao**, Mohammed Aldeen, Jingwen Yan, Long Cheng, Xiapu Luo, Haipeng Cai, Hongxin Hu, “Understanding GDPR Non-Compliance in Privacy Policies of Alexa Skills in European Marketplaces”, *The Web Conference (WWW)*, 2024, Acceptance rate: $406/2008 = 20\%$.
- [14] Wenbo Ding, **Song Liao**, Long Cheng, Ziming Zhao, and Hongxin Hu, “Command Hijacking on Voice-Controlled IoT in Amazon Alexa Platform”, *ACM ASIA Conference on Computer and Communications Security (ASIACCS)*, 2024, Acceptance rate: $55/284 = 19\%$.
- [15] **Song Liao**, Long Cheng, Haipeng Cai, Linke Guo, and Hongxin Hu, “SkillScanner: Detecting Policy-Violating Voice Applications Through Static Analysis at the Development Phase”, In *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2023, Acceptance rate: $235/1222 = 19\%$.
- [16] **Song Liao**, Ebuka Okpala, Long Cheng, Nishant Vishwamitra, Mingqi Li, Hongxin Hu, Feng Luo, and Matthew Costello, “Characterizing Offensive Tweets in the Era of COVID-19”, In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2023, Acceptance rate: $184/725 = 25\%$.
- [17] Wenbo Ding, **Song Liao**, Long Cheng, Ziming Zhao, Keyan Guo, and Hongxin Hu, “Exploring Vulnerabilities in Voice Command Skills for Connected Vehicles”, *EAI International Conference on Security and Privacy in Cyber-Physical Systems and Smart Vehicles (SmartSP)*, 2023.
- [18] Nishant Vishwamitra, Keyan Guo, **Song Liao**, Jaden Mu, Zheyuan Ma, Long Cheng, Ziming Zhao and Hongxin Hu, “Understanding and Analyzing COVID-19-related Online Hate Propagation Through Hateful Memes Shared on Twitter”, *The 2023 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*, 2023.
- [19] Matthew Costello, Nishant Vishwamitra, **Song Liao**, Long Cheng, Feng Luo, and Hongxin Hu, “COVID-19 and Sino-phobia: Detecting Warning Signs of Radicalization on Twitter and Reddit”, *Cyberpsychology, Behavior, and Social Networking*, 2023.
- [20] Jeffrey Young*, **Song Liao***, Long Cheng, Hongxin Hu, and Huixing Deng, “SkillDetective: Automated Policy-Violation Detection of Voice Assistant Applications in the Wild”, In *31st USENIX Security Symposium (USENIX Security)*, 2022, Acceptance rate: $256/1414 = 18\%$.
- [21] **Song Liao**, Christin Wilson, Long Cheng, Hongxin Hu, and Huixing Deng, “Problematic Privacy Policies of Voice Assistant Applications”, *IEEE Security & Privacy Magazine*, 2021.
- [22] Mingqi Li, **Song Liao**, Ebuka Okpala, Max Tong, Matthew Costello, Long Cheng, Hongxin Hu, and Feng Luo, “COVID-HateBERT: a Pre-trained Language Model for COVID-19 related Hate Speech Detection”, In *2021 20th IEEE International Conference on Machine Learning and Applications (ICMLA)*, 2021.
- [23] Matthew Costello, Long Cheng, Feng Luo, Hongxin Hu, **Song Liao**, Nishant Vishwamitra, Mingqi Li, and Ebuka Okpala, “COVID-19: a pandemic of Anti-Asian cyberhate”, *Journal of Hate Studies*, 17(1), 2021.
- [24] **Song Liao***, Christin Wilson*, Long Cheng, Hongxin Hu, and Huixing Deng, “Measuring the effectiveness of privacy policies for voice assistant applications”, In *Annual Computer Security Applications Conference (ACSAC)*, 2020, Acceptance rate: $70/302 = 23\%$, **Distinguish Paper Award**.
- [25] Long Cheng, Christin Wilson, **Song Liao**, Jeffrey Young, Daniel Dong, and Hongxin Hu, “Dangerous skills got certified: Measuring the trust-worthiness of skill certification in voice personal assistant platforms”, In *Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2020, Acceptance rate: $121/715 = 17\%$.

WORKSHOP

- [1] **Song Liao**, Jingwen Yan, Yichen Liu, David Kotz, Luyi Xing, Long Cheng, “Towards Ubiquitous and Automated User Privacy Configuration”, *NDSS Workshop on Security and Privacy in Standardized IoT (SDIoTSec)*, 2026.
- [2] Jingwen Yan, Mohammed Aldeen, Jalil Harris, Kellen Grossenbacher, **Song Liao**, Feng Luo, Long Cheng, “Understand-

ing User Acceptance of Privacy Labels: Barriers and Enhancements”, Proceedings of the 24th Workshop on Privacy in the Electronic Society (WPES), 2025

- [3] Yichen Liu, Jingwen Yan, **Song Liao**, Long Cheng, Luyi Xing, “WIP: Towards Privacy Compliance by Design in the Matter Protocol”, NDSS Workshop on Security and Privacy in Standardized IoT (SDIoTSec), 2025.
- [4] **Song Liao**, Jingwen Yan, Long Cheng, “WIP: Hidden Hub Eavesdropping Attack in Matter-enabled Smart Home Systems”, The NDSS Workshop on Security and Privacy in Standardized IoT (SDIoTSec), 2024.

AWARDS

- Clemson Outstanding Ph.D. Student in Computer Science, 2024
- Distinguished Paper Award, Annual Computer Security Applications Conference (ACSAC), 2020
- Google awarded us \$5,000 bug bounty for discovering vulnerabilities in Google Actions.
- Travel grant from CCS (2023), SmartSP (2023), IEEE SecDev (2022)

MEDIA AND NEWS

- We presented our work on Policy Violation Detection of Voice Assistant Applications at the FTC’s PrivacyCon 2021.
- Google awarded us two bug bounties for reporting various policy-violating Google Actions (2021).
- We presented our work on Trustworthiness of Skill Certification in Voice Personal Assistant Platforms at the FTC’s PrivacyCon 2020.
- The Register, Washington Internet Daily, ZDNet and some other media reported our work on measuring the trustworthiness of Alexa’s skills certification (2020).
- The Register and VentureBeat reported our work on Alexa skill privacy policy analysis (2020).

TEACHING

Lecturer

Texas Tech University:

- CS-4392 Computer Networks, Fall 2026
- CS-5340 Introduction to Information and Computer Security, 63 graduate students, eval: 4.67/5, Spring 2026
- CS-4331 Special Problems in Computer Science: Applied Data Science, 38 undergraduate students, eval: 4.39/5, Fall 2025
- CS-5331 Special Problems in Computer Science: Applied Data Science, 52 graduate students, eval: 4.81/5, Fall 2025
- CS-5120 Computer Science Graduate Seminar, 79 graduate students, eval: 4.5/5, Fall 2025
- CS-4392 Computer Networks, 47 undergraduate students, eval: 4.1/5, Spring 2025
- CS-5331 Special Problems in Computer Science: Applied Data Science, 30 undergraduate students, eval: 4.5/5, Fall 2024

Clemson University:

- CPSC-6200/4200 Computer Security Principles, Summer 2024

Graduate Teaching Assistant (TA)

Clemson University:

- CPSC-6200/4200 Computer Security Principles, Fall 2023
- CPSC-8570 Network Technologies Security, Spring 2023
- CPSC-6200/4200 Computer Security Principles, Fall 2022
- CPSC-6200/4200 Computer Security Principles, Summer 2022
- CPSC-6200/4200 Computer Security Principles, Spring 2021

TALKS

- “Ensuring the Privacy Compliance of Voice Personal Assistant Applications”, University of Delaware, Virtual, 2026
- “Ensuring the Privacy Compliance of Voice Personal Assistant Applications”, University of North Texas, Virtual, 2025
- “A First Look at Security and Privacy Risks in the RapidAPI Ecosystem”, CCS, Salt Lake City, 2024
- “SkillScanner: Detecting Policy-Violating Voice Applications Through Static Analysis at the Development Phase”, CCS, Copenhagen, Denmark, 2023
- “Exploring Vulnerabilities in Voice Command Skills for Connected Vehicles”, SmartSP, Chicago, 2023
- “Measuring the effectiveness of privacy policies for voice assistant applications”, ACSAC, Virtual, 2020

PROFESSIONAL SERVICE

Technical Program Committee Member:

- USENIX Security Symposium (Usenix Security), 2027
- Network and Distributed System Security Symposium (NDSS), 2026, 2027

- The Web Conference (WWW), 2025, 2026
- Annual Computer Security Applications Conference (ACSAC), 2025, 2026
- International Conference on Advances in Social Networks Analysis and Mining (ASONAM), 2025
- EAI International Conference on Security and Privacy in Cyber-Physical Systems and Smart Vehicles (SmartSP), 2025, 2026
- IEEE TPS 2026 (IEEE International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS)), 2026

UNIVERSITY SERVICE

Department Committees:

- Member, Graduate Admissions Committee
- Member, Graduate Program Committee

PAPER REVIEW

- ACM Conference on Computer and Communications Security (CCS), 2024
- The Web Conference (WWW), 2024, 2025, 2026
- Transactions on Information Forensics & Security (TIFS), 2025, 2024
- IEEE/ACM Transactions on Networking (ToN), 2024, 2023
- Computers & Security, 2024
- IEEE Transactions on Dependable and Secure Computing (TDSC), 2025, 2023
- IEEE/ACM International Conference on Automated Software Engineering (ASE), 2023
- International Journal of Human-Computer Interaction (IJHCI), 2023
- Annual Computer Security Applications Conference (ACSAC), 2023, 2022, 2021, 2020
- IEEE Transactions on Reliability, 2023
- International Conference on Distributed Computing Systems (ICDCS), 2025, 2023
- International Conference on Computer Communications and Networks (ICCCN), 2023
- International Conference on Information and Communications Security (ICICS), 2023
- International Conference on Security and Privacy in Communication Networks (SecureComm), 2023
- Cyberpsychology, Behavior, and Social Networking, 2023
- IEEE International Conference on Computer Communications (INFOCOM), 2022, 2021
- IEEE International Conference on Parallel and Distributed Systems (ICPADS), 2022
- IEEE Conference on Communications and Network Security (CNS), 2021, 2020
- International Performance Computing and Communications Conference (IPCCC), 2021, 2020
- IEEE Global Communications Conference (GLOBECOM), 2020, 2019
- IEEE International Conference on Internet of Things (iThings), 2019

STUDENTS

Ph.D. students:

- Weiliang Chen
- Huan Zhao

Graduate students:

- Felix Sarpong
- Salish Kumar
- Farzana Fariha
- Aurore Munyaneza
- Lirong Chen

Undergraduate students:

- Elena Maccormack

Dissertation/Thesis Committee Member

- Zhenyu Xu (PhD)
- Ruoyu Xu (PhD)
- Sarpong Felix (MS, chair)
- Salish Kumar (MS, chair)
- Sahil Babu Bhetwal (MS)

COLLABORATORS

- Hongxin Hu, Professor at the University at Buffalo
- Haipeng Cai, Associate Professor at the University at Buffalo
- Luyi Xing, University of Illinois Urbana-Champaign
- Xiapu Luo, Professor at The Hong Kong Polytechnic University
- Danfeng (Daphne) Yao, Professor at North Carolina State University
- Feng Luo, Professor at Clemson University
- Nishant Vishwamitra, Assistant Professor at the University of Texas at San Antonio
- Matthew Costello, Associate Professor at Clemson University